

Algebra II Exponential and Logarithmic Cheatsheet

Module 7

Inverse Definitions

$$y = b^x \iff x = \log_b y$$
$$\log_b b^x = x, \quad b^{\log_b x} = x$$

Conditions:

$$b > 0, \quad b \neq 1, \quad x > 0$$

Exponent Rules

$$b^m \cdot b^n = b^{m+n}$$
$$\frac{b^m}{b^n} = b^{m-n}$$
$$(b^m)^n = b^{mn}$$
$$b^{-n} = \frac{1}{b^n}, \quad b^0 = 1$$

Log Rules

$$\log_b(MN) = \log_b M + \log_b N$$
$$\log_b \left(\frac{M}{N} \right) = \log_b M - \log_b N$$
$$\log_b(M^p) = p \log_b M$$
$$\log_b a = \frac{\log a}{\log b} = \frac{\ln a}{\ln b}$$

Solving Templates

Same base:

$$b^{f(x)} = b^{g(x)} \Rightarrow f(x) = g(x)$$

One log:

$$\log_b(f(x)) = k \Rightarrow f(x) = b^k$$

Log equals log:

$$\log_b(f(x)) = \log_b(g(x)) \Rightarrow f(x) = g(x)$$

Check:

$$f(x) > 0, \quad g(x) > 0$$

Graph Features

$$y = ab^{x-h} + k$$

Horizontal asymptote:

$$y = k$$

$$y = a \log_b(x - h) + k$$

Vertical asymptote:

$$x = h$$

Growth and Decay

$$y = ab^t, \quad y = ae^{kt}$$

$$k > 0 \Rightarrow \text{growth}, \quad k < 0 \Rightarrow \text{decay}$$

$$\text{doubling time} = \frac{\ln 2}{k}$$

$$\text{half-life} = \frac{\ln(1/2)}{k}$$

Quick Checks

- Log arguments must be positive.
- Exponential outputs are positive before vertical shifts.
- Do not distribute logs over addition.
- Use change of base for calculator evaluation.